A Reinforcement Learning Framework for Dynamic Task Prioritization and Workflow Routing in Medical Imaging Systems

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Table of Contents

Abstract

Medical imaging departments are experiencing significant operational challenges due to increasing imaging volumes, limited radiologist availability, inefficient task prioritization, and prolonged reporting turnaround times. Conventional workflow management approaches, including first-in-first-out scheduling and static rule-based prioritization, are unable to adapt effectively to dynamic operational conditions such as fluctuating queue congestion, varying study urgency, modality diversity, and uneven radiologist workload distribution. These limitations may negatively affect diagnostic efficiency, urgent case handling, resource utilization, and overall patient care quality. The purpose of this research is to develop an intelligent reinforcement learning-based framework for adaptive task prioritization and workflow routing in medical imaging systems. The proposed study models radiology workflow optimization as a sequential decision-making problem using a Markov Decision Process (MDP). A Proximal Policy Optimization (PPO) agent will be designed to learn adaptive scheduling and routing policies based on operational workflow states, including queue status, urgency level, imaging modality, reporting deadlines, and radiologist workload. The framework further incorporates a multi-objective reward mechanism to optimize turnaround time, urgent case prioritization, and workload balancing simultaneously. A simulation-based radiology workflow environment will be developed using workflow distributions derived from publicly available datasets, including MIMIC-IV and MIMIC-CXR. The expected outcome of this study is an intelligent workflow optimization framework capable of outperforming traditional heuristic scheduling approaches in operational efficiency and urgent case management. The proposed research is expected to contribute to healthcare operational optimization by extending reinforcement learning applications beyond medical image interpretation into adaptive radiology workflow management.

1. Background

Medical imaging has become an essential component of modern healthcare systems for disease diagnosis, treatment planning, emergency response, and longitudinal patient monitoring [1]. Radiology departments routinely process large volumes of imaging studies generated from modalities such as computed tomography (CT), magnetic resonance imaging (MRI), ultrasound, and digital radiography. The continuous increase in imaging demand has created substantial operational pressure on healthcare organizations worldwide [2].

Radiology workflow management involves multiple interconnected operational stages, including imaging order placement, image acquisition, worklist prioritization, radiologist assignment, reporting, and clinical communication [3]. Despite advancements in healthcare digitization, many radiology departments continue to depend on static scheduling approaches and manually prioritized worklists. Conventional scheduling strategies such as first-in-first-

The growing shortage of radiologists, increasing study complexity, and rising expectations for rapid reporting have further intensified workflow inefficiencies in medical imaging systems [5]. Delayed interpretation of urgent imaging studies may adversely affect emergency diagnosis, treatment planning, and clinical outcomes. In addition, uneven workload distribution may contribute to radiologist fatigue, reduced operational efficiency, and increased reporting variability [6].

Recent developments in artificial intelligence and reinforcement learning have demonstrated strong capabilities in solving sequential decision-making and operational optimization problems [7]. Reinforcement learning enables intelligent agents to learn adaptive decision-making policies through continuous interaction with complex environments. Unlike conventional optimization methods, reinforcement learning algorithms can dynamically adapt workflow decisions based on changing operational states and delayed reward mechanisms [8].

Although artificial intelligence research in radiology has predominantly focused on image analysis, lesion detection, segmentation, and disease classification, limited research has explored operational workflow optimization after image acquisition [9]. Consequently, there is a growing need for intelligent workflow orchestration systems capable of dynamically prioritizing imaging tasks, balancing radiologist workload, and optimizing turnaround time under real-time operational constraints.

This research proposes a reinforcement learning-based framework for adaptive task prioritization and workflow routing in medical imaging systems. The proposed framework integrates simulation-based workflow optimization, multi-objective reward mechanisms, and healthcare interoperability concepts through HL7-aware workflow ingestion and PACS integration architecture. The study seeks to improve operational efficiency, urgent case handling, and radiologist workload balancing within modern radiology environments.

Domain	**Existing Methods**	**Key Limitation**	**Research Gap**	**Proposed Solution**
Radiology Workflow	FIFO scheduling	Static prioritization	No adaptive learning	PPO-based workflow RL
Queue Optimization	Heuristic methods	Poor scalability	Limited sequential optimization	RL queue optimization
PACS Workflow	Conventional RIS/PACS	Passive workflow handling	No intelligent orchestration	RL-integrated PACS
Radiologist Assignment	Manual allocation	Workload imbalance	Lack of optimization intelligence	RL-based assignment strategy

2. Literature Review

The integration of artificial intelligence into healthcare systems has significantly expanded across clinical decision support, operational analytics, and medical image analysis applications [10]. Among various machine learning paradigms, reinforcement learning (RL) has emerged as a promising approach for solving sequential decision-making problems in dynamic and uncertain environments [11]. RL enables intelligent agents to learn optimal policies through continuous interaction with operational systems using reward-driven optimization mechanisms.

Several studies have investigated reinforcement learning applications in healthcare operations management. Komorowski et al. [12] applied reinforcement learning for treatment policy optimization in intensive care environments, demonstrating the capability of RL to manage complex sequential healthcare decisions. Yu et al. [13] explored deep reinforcement learning for hospital resource scheduling and reported improvements in operational efficiency and resource allocation. Similarly, RL-based optimization has been studied for emergency department management, hospital bed allocation, and patient flow optimization [14]. These studies indicate that RL-based systems can outperform traditional heuristic and rule-based scheduling approaches in highly dynamic operational settings.

Traditional healthcare workflow optimization methods commonly rely on operations research, queueing theory, discrete-event simulation, and mathematical optimization techniques [15]. While these approaches provide structured optimization capabilities, they often lack real-time adaptability and struggle to manage continuously changing workflow conditions. Static scheduling systems generally fail to incorporate dynamic operational variables such as fluctuating workload, emergency prioritization, reporting delays, and resource availability.

Within the medical imaging domain, existing artificial intelligence research has largely concentrated on image interpretation tasks, including lesion detection, segmentation, disease classification, and computer-aided diagnosis [16]. Rajpurkar et al. [17] demonstrated the application of deep convolutional neural networks for chest radiograph interpretation using the CheXNet framework. Similarly, numerous studies have investigated AI-assisted image classification and radiological decision support systems [18]. However, relatively limited attention has been directed toward post-acquisition workflow optimization and operational orchestration within radiology departments.

Current radiology workflow systems typically utilize static prioritization strategies such as first-in-first-out scheduling, manually escalated urgent cases, or fixed rule-based queues [19]. Although these approaches provide operational simplicity, they lack intelligent adaptability to changing queue congestion, radiologist workload imbalance, and imaging urgency variability. Furthermore, existing workflow optimization studies frequently optimize a single operational objective, such as turnaround time reduction, without simultaneously addressing workload fairness, urgent case handling, and operational efficiency [20].

Several research gaps remain evident within the current literature. First, limited studies have explored reinforcement learning for dynamic radiology workflow routing and task prioritization. Second, minimal research has integrated healthcare interoperability standards such as HL7, RIS, and PACS into RL-based workflow architectures. Third, insufficient attention has been given to multi-objective reward optimization balancing turnaround time, workload fairness, urgent case prioritization, and operational stability.

The proposed study addresses these limitations by introducing a reinforcement learning framework for adaptive radiology workflow optimization. The framework incorporates multi-objective reward optimization, simulation-driven operational modeling, HL7-aware workflow ingestion, dynamic radiologist assignment, case prioritization and intelligent workflow routing mechanisms. Unlike prior studies primarily focused on image interpretation, this research extends reinforcement learning applications into operational imaging workflow management with strong practical relevance for modern healthcare systems.

Table:

Sl No	Title	Link to the Paper with year	Understanding of the Dataset	Understanding the Methodology Used	Evaluation Matrix	Research Gap
1	Operational optimization in radiology: An intelligent information management platform significantly enhances workflow efficiency and user satisfaction	https://www.sciencedirect.com/science/article/pii/S1687850725005047 July 2025	Hybrid dataset combining real, synthetic, and augmented images with segmentation labels and quality metrics.	Uses retrospective data analysis with systematic extraction, filtering, and statistical evaluation of key variables.	Accuracy, Dice score, sensitivity, specificity, AUC.	Limited generalization, explainability, and real-world clinical validation.
2	Radiology Workflow Assistance With Artificial Intelligence:	https://www.sciencedirect.com/science/article/pii/S154	Aggregated literature-derived datasets of AI	Framework-based review analyzing AI across radiology	Workflow efficiency, turnaround time, resource	Isolated workflow optimization lacking integration

	Establishing the Link to Outcomes	6144025005988 October 2025	applications and performance metrics across radiology workflow stages.	workflow stages, evaluating efficiency, accuracy, and clinical outcome impact.	utilization, task completion rate.	and patient outcome validation.
3	Optimization of medical radiation technologist schedules using advanced analytical tools	https://www.sciencedirect.com/science/article/pii/S1939865425001493 June 2025	Real-world staffing and operational data including schedules, demand patterns, constraints, and workforce availability.	Develops cost-constrained optimization model using real staffing data, applying constraints to generate efficient schedules.	Accuracy, precision, recall, F1-score, AUC.	Lacks adaptive learning, multimodal integration, and real-world deployment.
4	Automation of medical imaging business reporting workflows in Ontario for quantitative and qualitative process improvement	https://www.sciencedirect.com/science/article/pii/S1939865425000414 April 2025	Operational workflow data including task times, resources, constraints, and system usage for optimization modelling.	Applies two-phase optimization approach to automate workflows, improving efficiency, time savings, and user experience.	Accuracy, latency, scalability, system performance metrics.	Limited interoperability, scalability, and validation across diverse clinical environments.
5	The Impact of AI on Workflow Optimization in Medical Imaging Departments	https://www.researchgate.net/publication/390114120_	Demonstration-based web interaction dataset	Uses workflow-guided reinforcement learning,	Efficiency improvement, turnaround time, error	Insufficient large-scale validation and standardize

		The_Impact_of_AI_on_Workflow_Optimization_in_Medical_Imaging_Departments	with structured workflows, actions, and rewards for training RL agents.	constraining exploration with demonstrations to improve task efficiency and learning performance.	reduction, user satisfaction.	d real-world workflow integration.

Cumulative Research Gaps

✓ **Lack of End-to-End System Integration**

- Most studies focus on isolated tasks (segmentation, scheduling, reporting)
- No unified full radiology workflow system

✓ **Limited Real-World Clinical Validation**

- Evaluations are mostly lab-based or single-center
- Missing multi-center, large-scale deployment evidence

✓ **Poor Interoperability with Clinical Infrastructure**

- Weak integration with:
  - PACS
  - EHR
  - Hospital IT systems
- Lack of standardized deployment frameworks

✓ **Absence of Adaptive / Learning Systems**

- AI models are static
- No use of:
  - Reinforcement Learning
  - Continuous feedback from clinicians

✓ **Limited Impact on Patient-Centric Outcomes**

- Focus on:
  - Accuracy
  - Efficiency
- Little evidence on:
  - Patient outcomes
  - Clinical decision improvement

✓ **Lack of explainability and Trustworthiness**

- Models behave as black boxes
- Limited use of:
  - Explainable AI (XAI)
  - Clinician-interpretable outputs

✓ **No Uncertainty Quantification**

- Predictions are deterministic
- No confidence/risk-aware decision support

✓ **Limited Multi-Modal Data Integration**

- Mostly single modality (e.g., only imaging)
- Missing integration of:
  - Clinical records
  - Reports
  - Multi-modal imaging

✓ **Dataset Limitations**

- Small or curated datasets
- Lack of:
  - Diverse populations
  - Robust generalization

- ✓ **Missing Production-Grade Deployment Considerations**

- No focus on:
  - Scalability
  - Privacy (HIPAA/GDPR)
  - Monitoring and auditability

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3. Research Questions

The proposed study seeks to address the following research questions:

1. How can reinforcement learning be applied to dynamically prioritize imaging tasks within radiology workflow systems?
2. Can reinforcement learning-based workflow routing reduce imaging turnaround time compared with traditional heuristic scheduling approaches?
3. How effectively can reinforcement learning balance radiologist workload while maintaining urgent case prioritization?
4. What operational improvements can be achieved using multi-objective reward optimization in medical imaging workflows?
5. How can healthcare interoperability standards such as HL7 and RIS integration be incorporated into reinforcement learning-based workflow optimization architectures?

4. Aim and Objectives

Aim

The primary aim of this research is to develop a reinforcement learning-based framework for adaptive task prioritization and workflow routing in medical imaging systems.

Objectives

1. To model radiology workflow optimization as a Markov Decision Process (MDP).
2. To design a multi-objective reward function incorporating turnaround time reduction, urgent case prioritization, and radiologist workload balancing.
3. To develop a simulation-based radiology workflow environment using publicly available healthcare datasets.
4. To implement a Proximal Policy Optimization (PPO)-based reinforcement learning agent for workflow optimization.
5. To integrate HL7-aware workflow ingestion concepts into the proposed architecture.
6. To evaluate the proposed RL framework against conventional scheduling approaches such as FIFO and rule-based prioritization.
7. To analyze the operational benefits of adaptive workflow routing in medical imaging systems.

5. Significance of the Study

This study is significant because it addresses operational inefficiencies in radiology workflow systems that directly influence healthcare delivery quality, reporting efficiency, and patient outcomes. Delays in imaging interpretation may adversely affect emergency diagnosis, treatment planning, and clinical decision-making processes.

The proposed reinforcement learning framework introduces adaptive workflow decision-making capabilities capable of dynamically responding to operational changes such as queue congestion, modality distribution, study urgency, and radiologist workload. By optimizing workflow routing and task prioritization, the proposed framework seeks to improve turnaround time, enhance urgent case management, and balance workload distribution among radiologists.

From an academic perspective, this research contributes to the growing field of AI-driven healthcare operations optimization by extending reinforcement learning applications beyond image interpretation into operational radiology workflow management. The study further contributes methodological innovation through multi-objective reward optimization and intelligent workflow orchestration.

From an industrial perspective, the proposed framework may support radiology departments, hospital imaging centers, healthcare technology vendors, PACS providers, and AI-enabled workflow management platforms. The integration of HL7-aware workflow ingestion and PACS interoperability concepts further enhances the practical deployment potential of the proposed system.

6. Scope of the Study

The scope of this study is limited to operational workflow optimization after image acquisition within radiology systems. The proposed research specifically focuses on:

- Imaging task prioritization
- Workflow routing optimization
- Queue management
- Radiologist workload balancing
- Reinforcement learning-based operational decision-making
- Simulation-based radiology workflow optimization

The study does not address:

- Medical image interpretation
- Disease diagnosis
- Image segmentation or reconstruction
- Clinical treatment decision-making
- Real-time hospital deployment

The proposed framework will be validated through simulation-based experimentation rather than live clinical deployment.

7. Research Methodology

This study adopts a quantitative and simulation-based research methodology for adaptive radiology workflow optimization. The proposed workflow optimization problem is formulated as a Markov Decision Process (MDP), where a reinforcement learning agent interacts continuously with a simulated radiology workflow environment to learn optimal operational policies.

Reinforcement Learning Agent Design: The proposed reinforcement learning framework models workflow optimization as a sequential decision-making problem.

Intelligent Agent Design

The proposed reinforcement learning framework is modeled as a learning agent capable of adaptive workflow optimization through continuous interaction with the operational environment.

Unlike conventional utility-based systems relying on static decision rules, the proposed agent dynamically learns workflow policies using reward-driven optimization. The agent continuously updates its decision strategy based on workflow states, queue congestion, urgency levels, radiologist workload, and operational outcomes.

The agent behavior is guided through a multi-objective utility function represented by the reinforcement learning reward mechanism. Consequently, the proposed architecture combines learning-based adaptation with utility-driven operational optimization.

State Representation

The workflow state at time t is represented as:

$S_t = [Q, U, L, M, D]$

Where:

Q = Queue status for imaging studies

- U = Study urgency level
- L = Radiologist workload distribution
- M = Imaging modality type
- D = Remaining reporting deadline

Example workflow state variables include:

- Number of pending CT studies
- Number of urgent MRI studies
- Current radiologist workload
- Estimated queue waiting time

Action Space

The reinforcement learning agent selects workflow routing and prioritization actions represented as:

A_t = {assign_r1, assign_r2, assign_r3, escalate}

Possible actions include:

- Assigning imaging studies to radiologists
- Prioritizing urgent imaging studies
- Escalating critical imaging tasks
- Dynamically balancing radiologist workload

Mathematical Formulation

Markov Decision Process:

M = (S, A, P, R, Ɣ)

Reward Function:

- M → The whole decision-making system (MDP)
- S → States / all possible situations (patient queue, scan status, urgency level)
- A → Actions , means what the agent can do. (prioritize scan, assign radiologist, delay case)
- **P** → Transition Probability , means P(s′|s,a)P(s' | s, a)P(s′|s,a) which is , probability of moving to next state s′ from state s after action a
- **R** → Reward Function which is feedback signal
- **γ** (gamma) → Discount Factor (0 to 1)→ How much future rewards matter
    - ➔ γ=0 → only care about immediate reward
    - ➔ γ=0.99\gamma = 0.99γ=0.99 → care about long-term outcomes

Reward Function Design section next is capturing the same

Objective Function:

J(∏)= E[∑ **γ**^t R_t]

This equation says: "Maximize total future rewards over time"

Symbol Interpretation:

- **J(π)** = Performance of policy π . How good your decision strategy is
- **π** = Policy with strategy (mapping from state → action)
- **E** = Expectation which is average over many runs
- **γ^t** = Discounting factor over time. Future rewards become less important
- **R_t** = Reward at time step t

PPO Objective Function:

L(**θ**) = E[min(r_t(**θ**)A_t, clip(r_t(**θ**),1- **ε**,1+ **ε**)A_t)]

This equation says: "Maximize total future rewards over time"

L(θ)→ Objective we optimize → Depends on parameters θ (neural network weights)

rt(θ) → Probability Ratio where, rt(θ)=πθold(at|st)πθ(at|st)

Meaning, how much the new policy changed vs old policy

At → Advantage Function

Measures how good an action is:

- Positive → good action
- Negative → bad action

clip(...) → Limits how much policy can change

- Range: 1−ϵ to 1+ϵ1
- ϵ is small (like 0.1 or 0.2)

This Prevents:

- Too large updates
- Training instability

PPO is basically saying: "Improve policy—but don't change it too much at once"

Reinforcement Learning Model

The proposed framework utilizes the Proximal Policy Optimization (PPO) reinforcement learning algorithm because of its stable policy optimization capability, efficient convergence behavior, and suitability for dynamic multi-objective operational environments [5], [8]. PPO has demonstrated strong performance in continuous optimization tasks and sequential operational decision-making problems.

The PPO agent learns adaptive workflow policies by maximizing cumulative rewards across simulated workflow episodes. The agent continuously updates workflow routing strategies based on operational changes, including queue congestion, urgency fluctuations, modality distribution, and radiologist workload.

Architecture should use a **Learning Agent with utility-based optimization behavior**. More precisely, the proposed RL framework is a learning agent architecture implementing utility-driven sequential decision optimization.

System is a learning agent because the agent -

- observes environment states
- learns from rewards
- improves policy over time
- adapts dynamically
- updates decisions continuously
- It improves behavior through interaction.

Reward Function Design

The proposed framework employs a multi-objective reward function designed to optimize operational efficiency while maintaining the handling of clinical urgency.

The reward function is represented as

$Rt = -\alpha T + \beta U - \gamma\sigma(L)$

Where:

- T = Imaging turnaround time
- U = Urgent case completion indicator
- $\sigma(L)$ = Standard deviation of radiologist workload
- α, β, γ = Weight coefficients

Reward Objectives

Turnaround Time Minimization
The RL agent was penalized for increased reporting delays.

Urgent Case Prioritization

Positive rewards were assigned when urgent imaging studies satisfied the service-level targets.

Workload Balancing

The reward function penalizes uneven workload distributions among radiologists.

The proposed reward mechanism enables the RL agent to learn adaptive workflow policies that balance operational efficiency and clinical urgency.

The agent tries to maximize:

- lower turnaround time
- higher urgent case success
- balanced workload

System Architecture

The proposed system architecture integrates a medical imaging infrastructure, Picture Archiving and Communication Systems (PACS), reinforcement learning–based workflow optimization, and AI-assisted inference pipelines.
The architecture is designed to optimize imaging workflow operations, beginning with image ingestion through radiologist assignment and operational feedback optimization. The architecture contains: Healthcare Integration Layer, Imaging Infrastructure Layer, Workflow State Construction Layer, Reinforcement Learning Optimization Layer, AI Inference Orchestration Layer, Operational Feedback and Reward Layer.

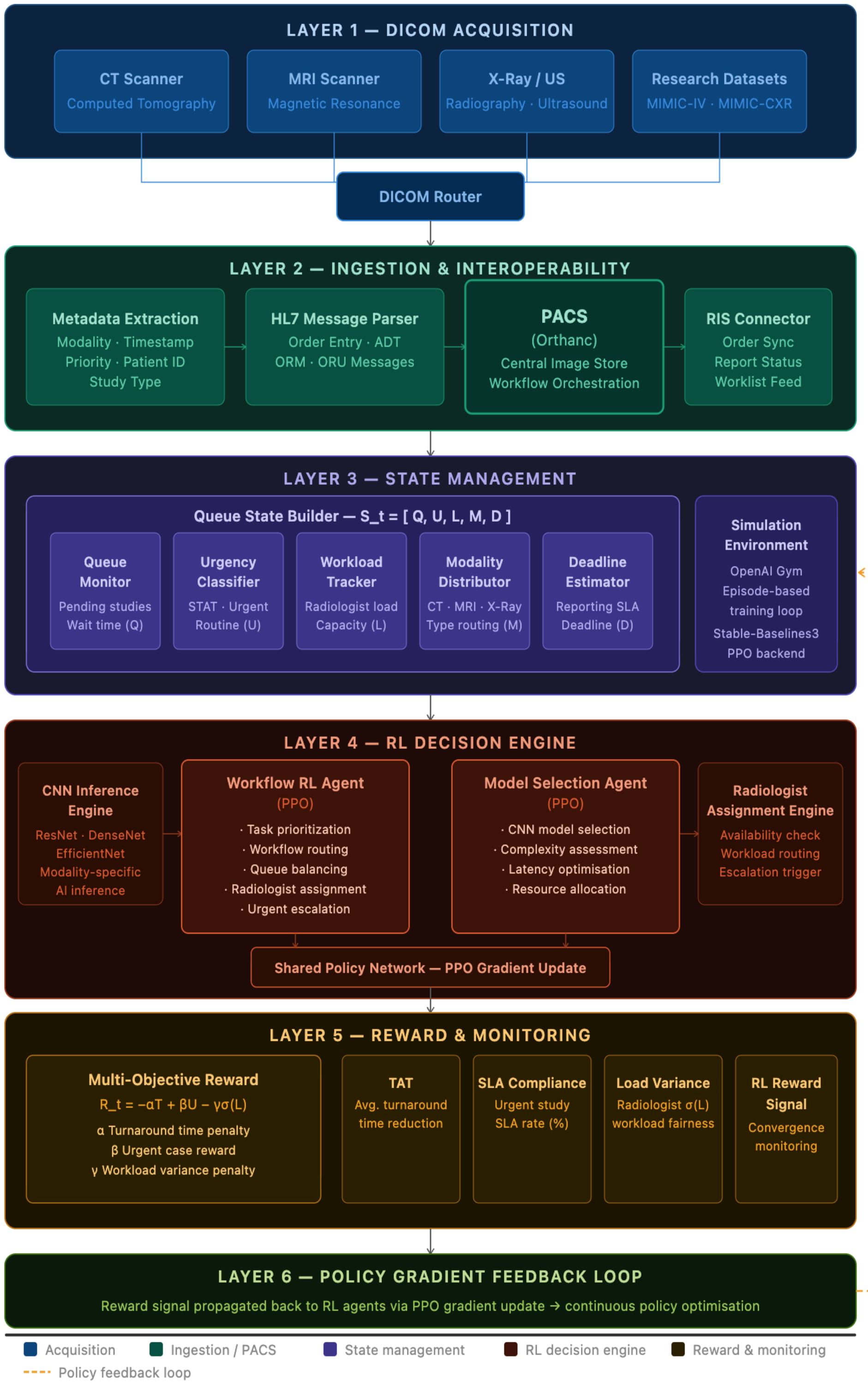


Figure 1a: System Architecture — RL Workflow Optimization Framework

- **Layer 1** (Acquisition) — Four distinct modality sources (CT, MRI, X-Ray/US, Research Datasets) converge through a single DICOM Router, matching your update that all data enters via PACS
- **Layer 2** (Ingestion & Interoperability) — Metadata Extraction Engine — reads the DICOM header tags from every study as it arrives (patient ID, modality, timestamp, priority flag, study type). This is essential which is the raw input to your state vector S_t.
  HL7 Message Parser — parses HL7 v2 messages (ORM = order, ADT = admission/transfer/discharge, ORU = results).

  In a real hospital, this is how the ordering physician's request travels from the HIS/EHR to the imaging department. For your simulation, you can derive order information from MIMIC-IV data directly, so this can be simulated rather than implemented.

  PACS (Orthanc) — the central image store. All DICOM images land here after acquisition. It also acts as the workflow orchestrator — the queue your RL agent will observe and act upon. This is non-negotiable.

  RIS Connector — the Radiology Information System handles scheduling, billing, and reporting status. It pushes worklist data back to radiologists and receives report completion events. In a live clinical deployment this matters a lot. But in your simulation-based thesis, the worklist is already modelled inside PACS/the Gym environment, so the RIS Connector is essentially a real-world integration concern, not a research concern.

- **Layer 3** (State Management) — Queue State Builder with exactly 5 non-overlapping sub-components (Queue Monitor, Urgency Classifier, Workload Tracker, Modality Distributor, Deadline Estimator), each mapping to a distinct state variable in S_t = [Q, U, L, M, D]. The Simulation Environment (OpenAI Gym) sits alongside
- **Layer 4** (RL Decision Engine) — CNN Inference and Radiologist Assignment as external engines feeding the two PPO agents, which share a common policy network. Bullet points inside each agent are mutually exclusive with no overlap
- **Layer 5** (Reward & Monitoring) — Formal reward function $R_t = -\alpha T + \beta U - \gamma\sigma(L)$ plus 4 distinct evaluation metrics
- **Layer 6** (Feedback Loop) — Policy gradient update described in one clean line, with the dashed amber arc returning to Layer 3

Workflow Architecture Overview

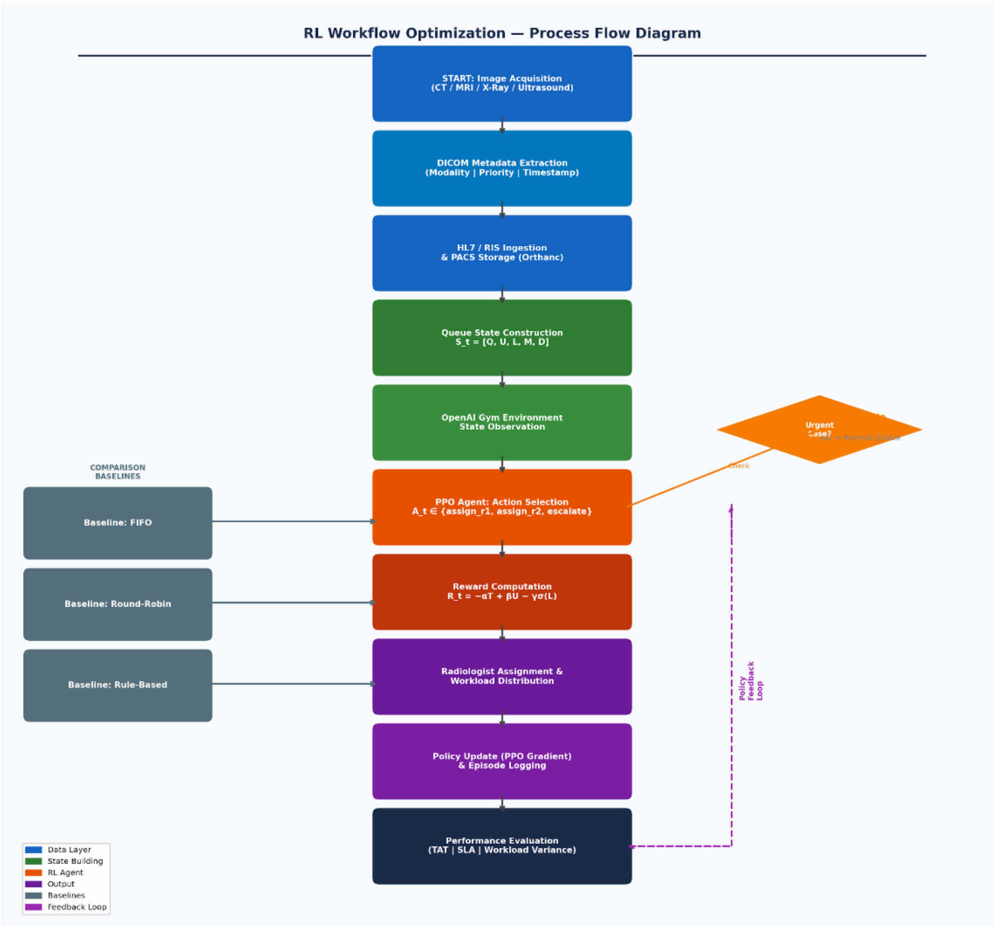


Figure 1b: RL Workflow Optimization — End-to-End Process Flow

The workflow begins with DICOM image acquisition from imaging modalities such as CT, MRI, and X-ray systems. Imaging metadata, including modality type, timestamp, patient priority, and examination details, were extracted during ingestion.

Imaging studies are subsequently ingested into a PACS platform, such as Orthanc, for centralized image management and workflow orchestration.

A queue state builder continuously monitors operational workflow conditions, including the following:

- Imaging queue congestion
- Study urgency
- Radiologist workload
- Imaging modality distribution
- AI inference backlog

These operational parameters were converted into structured workflow states for reinforcement learning processing.

The proposed framework incorporates two reinforcement learning agents:

Workflow RL Agent

The Workflow RL Agent is responsible for:

- Dynamic task prioritization
- Workflow routing optimization
- Queue balancing
- Radiologist assignment decisions

The agent learns adaptive scheduling policies that can optimize operational efficiency under changing workflow conditions.

Model Selection RL Agent

A secondary RL agent dynamically selects the most appropriate CNN inference model based on

- Imaging modality
- Study complexity
- Queue conditions
- Inference latency constraints
- Operational resource availability

This enables the intelligent orchestration of AI-assisted image analysis pipelines.

Following model selection, CNN inference was executed for AI-assisted preprocessing or prioritization support.

The workflow RL agent subsequently performs radiologist assignment based on the following:

- Current workload
- Study urgency
- AI inference outcomes

Reporting deadlines

Operational performance metrics, including turnaround time, queue waiting time, SLA compliance, and workload balancing, are continuously monitored for reward calculation and reinforcement learning optimization.

Architecture Components

DICOM Image Acquisition

Medical imaging modalities generate DICOM imaging studies containing imaging data and the associated metadata.

Metadata Extraction Layer

Workflow-relevant metadata are extracted from DICOM headers, including:

- Modality type
- Patient priority
- Acquisition timestamp
- Study identifiers
- PACS Integration Layer

Imaging studies were ingested into the Orthanc PACS for centralized storage and workflow coordination.

Queue State Builder

The queue state builder converts the operational workflow conditions into reinforcement learning state representations.

Workflow RL Agent

The workflow reinforcement learning agent dynamically optimizes the imaging task prioritization and routing.

Model Selection RL Agent

The model selection agent dynamically orchestrates AI inference workflows by selecting the optimal CNN models.

CNN Inference Engine
CNN-based inference models perform AI-assisted imaging analyses or workflow prioritization support. The MONAI [https://github.com/project-monai/monai] platform can be used as open source .

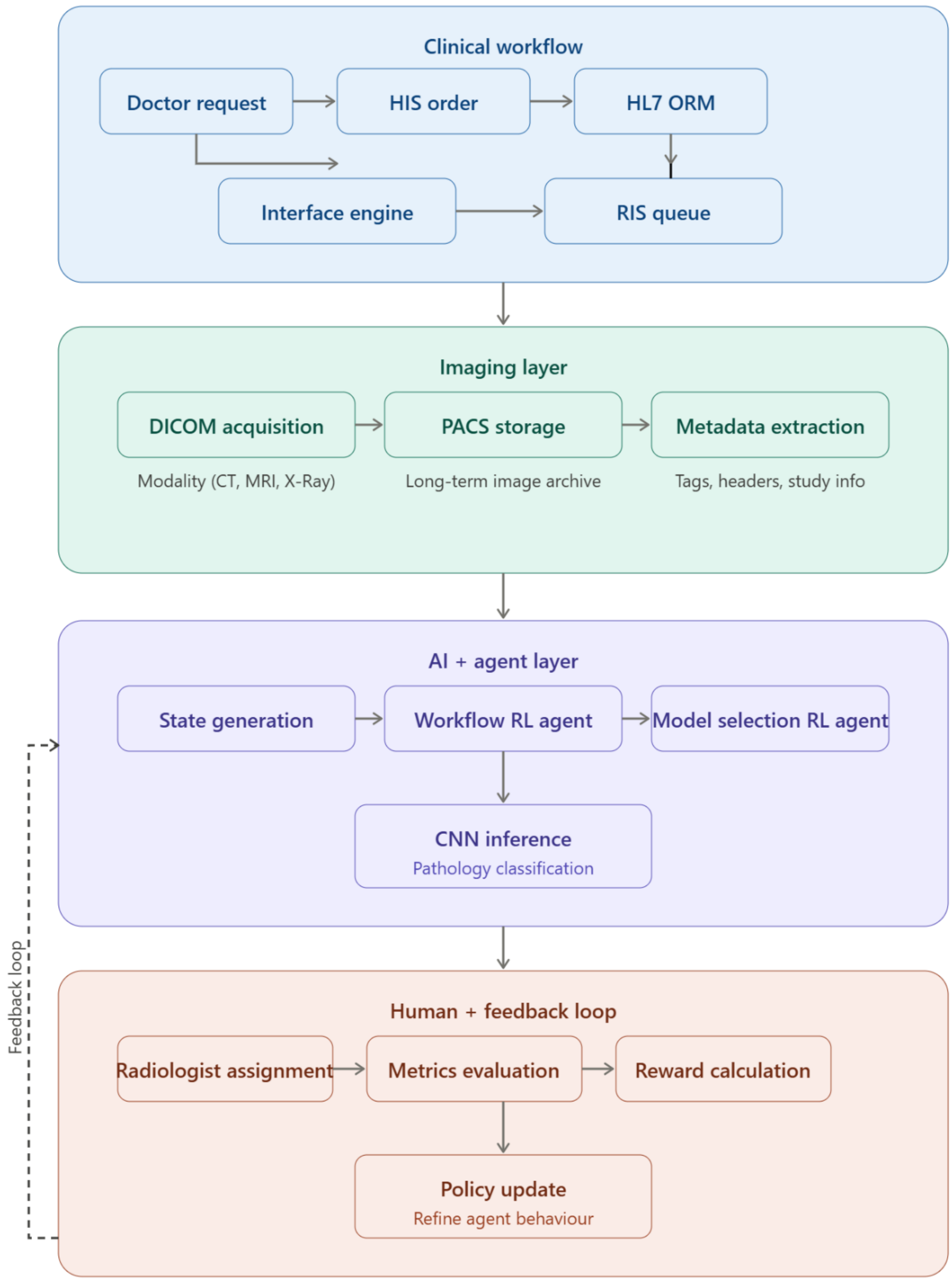
Radiologist Assignment Layer

The workflow RL agent assigns imaging studies to radiologists based on the operational optimization policies.

Reward Optimization Layer

The operational metrics were continuously evaluated for reinforcement learning reward computation.

Overall System Flow



Simulation Environment

A synthetic radiology workflow simulator was developed using Python to emulate operational imaging workflows.

The simulation environment will model the following:

- Imaging study arrivals
- Queue congestion
- Study urgency distribution
- Radiologist availability
- Reporting delays

Workflow distributions will be derived from publicly available healthcare datasets, including MIMIC-IV and MIMIC-CXR.

The simulation environment was implemented using the OpenAI Gym.

Evaluation Metrics

The proposed framework was evaluated using the following operational metrics:

Metric	**Purpose**
Average Turnaround Time	Measure reporting efficiency
SLA Compliance Rate	Evaluate urgent case handling
Queue Waiting Time	Measure workflow delay
Radiologist Load Variance	Evaluate workload balancing
RL Reward Convergence	Assess learning stability

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8. Requirement Resources

Software Requirements

- Python
- OpenAI Gym
- Stable-Baselines3
- Pandas
- NumPy
- Matplotlib

The system consists of:

- Frontend (React UI) / OHIFviewer
- Backend (FastAPI)
- Database (PostgreSQL for metadata)
- DICOM storage (Orthanc or local storage)
- RL agents (PPO-based)
- CNN inference models (simulated or MONAI-based)

Hardware Requirements

- MacBook Air M4 or equivalent system

Complete Dataset strategy

- MIMIC-IV Dataset [https://physionet.org/content/mimic-iv-fhir-demo/2.1.0/]
- MIMIC-CXR Dataset []
- Synthetic workflow simulation data [workflow metadata + operational event simulation]

because RL learns from:

- queue dynamics
- urgency
- workload
- timestamps
- routing decisions
- delays
- rewards

System needs 4 categories of data:

Purpose	Type
DICOM Images	CNN inference
Imaging Metadata	Workflow state
Operational Workflow Events	RL environment
Synthetic Queue Dynamics	Reward learning

Layer 1 — Medical Imaging Dataset

MIMIC-CXR [MIMIC-CXR Database v2.0.0] for chest X-rays

MIMIC-IV [https://physionet.org/content/mimiciv/3.1/]
https://physionet.org/content/mimic-iv-fhir-demo/2.1.0/

Can derive:

- modality
- patient priority approximation
- study timing
- reporting delays

Layer 2 — DICOM Metadata Dataset

Use metadata from:

DICOM Tag	Usage
StudyInstanceUID	study ID
Modality	CT/MR/XR
StudyDate	arrival time
AcquisitionTime	queue timing
BodyPartExamined	complexity
InstitutionName	site allocation

ProcedureCode	workflow type

Layer 3 — Workflow Event Simulation
No public hospital gives real RIS workflow logs. Simulating/Generating operational workflow is necessary .

Layer 4 — PACS Integration Dataset

Layer 5 — HL7 Workflow Dataset . Real HL7 datasets are difficult to obtain publicly. So simulating HL7 ORM messages is required for the simulation.

Layer 6 — RL Environment Dataset

Component	**Planned tool for approach**
Images	MIMIC-CXR
Metadata	DICOM headers
PACS	Orthanc
Workflow	Synthetic simulator
HL7	Simulated ORM messages
RL Environment	OpenAI Gym
Queue Engine	SimPy
RL Framework	Stable-Baselines3
CNN Models	PyTorch pretrained / MONAI Integration

9. Research Plan

Gantt Chart

Activities	Month 1	Month 2	Month 3	Month 4
Problem Formulation	■■			
Literature Review	■■	■		
Workflow Dataset Preparation	■■	■		
Simulation Environment Development		■■	■	
RL State-Action-Reward Design	■■	■		
PPO Agent Implementation		■■	■■	
Hyperparameter Optimization			■■	
Baseline Benchmarking			■■	
Experimental Evaluation			■	■■
Results Visualization				■■
Thesis Writing		■	■■	■■
Final Review and Submission				■■

Month 1

- Finalize research problem formulation
- Define RL state-action-reward structure
- Extract workflow distributions from public datasets
- Build synthetic workflow generator
- Conduct detailed literature review

Month 2

- Develop simulation environment
- Implement queue management logic
- Develop reward computation mechanism
- Implement baseline scheduling approaches

Month 3

- Train PPO reinforcement learning agent
- Optimize hyperparameters
- Conduct benchmarking experiments
- Analyze RL convergence behavior

Month 4

- Conduct final experimental evaluation
- Generate workflow performance visualizations
- Analyze operational results
- Complete thesis writing and documentation
- Final review and submission

Risks and Mitigation

Risk	Mitigation Strategy
Limited workflow data	Use synthetic simulation
RL training instability	Hyperparameter tuning
Time constraints	Focus on single RL algorithm
Integration complexity	Use simulated HL7 workflows

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